

11. Spinach is a leafy green vegetable.

- (a) Fertilisers containing copper(II) ethanoate are used to supply spinach with copper ions.

Copper(II) ethanoate can be made by reacting copper(II) carbonate with ethanoic acid.

- (i) Name the other products of this reaction.

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- (ii) Write the ionic formula of copper(II) ethanoate.

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- (b) Spinach is a source of oxalic acid.

A standard solution of oxalic acid can be used to determine the accurate concentration of a sodium hydroxide solution.

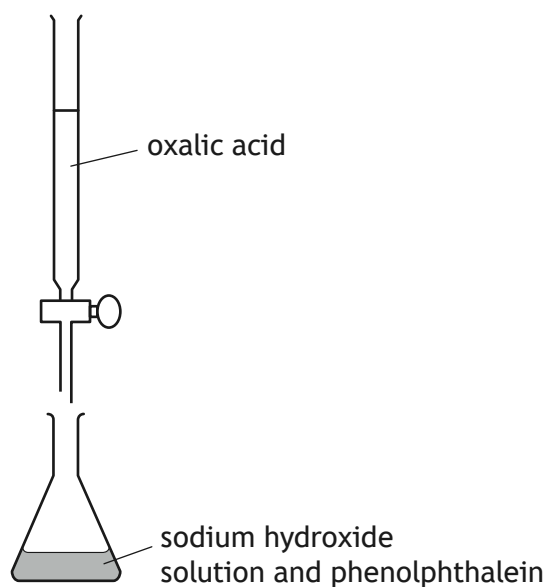
Given an accurately known mass of oxalic acid, describe **fully** how 250 cm³ of a standard solution of oxalic acid could be prepared.

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11. (continued)

- (c) The concentration of sodium hydroxide can be determined by titration with oxalic acid.



An accurate volume of sodium hydroxide solution is measured into a conical flask using a pipette.

- (i) Draw a diagram of a pipette suitable for measuring an accurate volume.

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- (ii) The indicator used in this titration is phenolphthalein. Phenolphthalein is colourless in acidic and neutral solutions but is pink in alkaline conditions.

State the colour change that would be observed at the end point in this titration.

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[Turn over



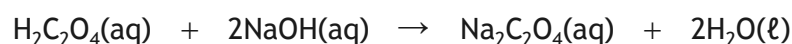
11. (c) (continued)

- (iii) The titration was repeated until results were obtained that were within 0.2 cm^3 of each other.

State the term used to describe titre volumes within 0.2 cm^3 of each other.

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- (d) The equation for the reaction of oxalic acid and sodium hydroxide is shown.



The concentration of sodium hydroxide solution was determined by titrating 25.0 cm^3 samples with 0.126 mol l^{-1} oxalic acid solution.

The average volume of oxalic acid solution required in the titration was 26.75 cm^3 .

Calculate the concentration, in mol l^{-1} , of the sodium hydroxide.

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[END OF QUESTION PAPER]

